

Introduction to Mathematics and Optimisation

Exercises Week 4

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1 Exercise 3.42

Let

$$D = \begin{pmatrix} d & 0 \\ 0 & e \end{pmatrix}$$

be a diagonal matrix. What conditions must the diagonal entries d and e satisfy in order for D to be positive definite?

Let

$$A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$$

denote a symmetric 2×2 matrix, where $a \neq 0$. Let

$$B = \begin{pmatrix} 1 & -\frac{c}{a} \\ 0 & 1 \end{pmatrix}.$$

Show that B is invertible and compute

$$B^\top AB.$$

Use this to show that A is positive definite if and only if $a > 0$ and $ab - c^2 > 0$.

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = 2x^2 + 3y^2 + 4xy.$$

Show that $f(x, y) \geq 0$ for every $x, y \in \mathbb{R}$.

1.1 Part 1

For the matrix D to be positive definite it $f_D(v) > 0$ must hold true for all nonzero v . For our matrix D the function $f_D(v)$ is defined as $dx^2 + ey^2$ so we must find conditions which guarantee that $dx^2 + ey^2 > 0$ is true for all nonzero vectors v .

As both x and y is raised to the second power we can disregard any negative values of x and y since $x^2 = (-x)^2$. We can also disregard the case where both x and y are equal to 0 because of the definition of positive definite. This means we only need to find the conditions for three cases:

- $\begin{pmatrix} x \\ 0 \end{pmatrix}$: This will result in the inequality $dx^2 > 0$ this raises the condition $d > 0$. x^2 is already positive and the product will be positive if and only if $d > 0$
- $\begin{pmatrix} 0 \\ y \end{pmatrix}$: This raises the condition $e > 0$ for the same reason as in the previous case
- $\begin{pmatrix} x \\ y \end{pmatrix}$: This does not raise any more conditions as the conditions $d > 0$ and $e > 0$ are enough for the inequality to always hold true.

So the conditions for the matrix D to be positive definite are $d > 0$ and $e > 0$

1.2 Part 2

We can show that B is invertible by calculating it's determinant and checking it's not equal to 0. In case of B the determinant is $1 * 1 + 0 * (-\frac{c}{a})$ which is equal to 1 which is not equal to 0 so matrix B is invertible.

Alternatively we can show that B is invertible by finding inverted matrix B^{-1} :

$$B = \begin{pmatrix} 1 & -\frac{c}{a} \\ 0 & 1 \end{pmatrix} \quad B^{-1} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}$$

We solve $BB^{-1} = I$:

$$\begin{pmatrix} 1 & -\frac{c}{a} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Multiplying out gives the system of four linear equations:

$$\begin{aligned} x_{11} - \frac{c}{a}x_{21} &= 1 \\ x_{12} - \frac{c}{a}x_{22} &= 0 \\ x_{21} &= 0 \\ x_{22} &= 1 \end{aligned}$$

From the third and fourth equations, we get:

$$x_{21} = 0, \quad x_{22} = 1$$

Substitute into the first two:

$$x_{11} = 1, \quad x_{12} = \frac{c}{a}$$

Hence,

$$B^{-1} = \begin{pmatrix} 1 & \frac{c}{a} \\ 0 & 1 \end{pmatrix}$$

Now we need to calculate $B^{\top}AB$:

$$\begin{aligned} \begin{pmatrix} 1 & 0 \\ -\frac{c}{a} & 1 \end{pmatrix} \begin{pmatrix} a & c \\ c & b \end{pmatrix} \begin{pmatrix} 1 & -\frac{c}{a} \\ 0 & 1 \end{pmatrix} &= \\ \begin{pmatrix} 1 & 0 \\ -\frac{c}{a} & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ c & b - \frac{c^2}{a} \end{pmatrix} &= \\ \begin{pmatrix} a & 0 \\ 0 & b - \frac{c^2}{a} \end{pmatrix} & \end{aligned}$$

Now as per proposition 3.41 since A is symmetric and B is invertible A is positive definite if and only if the resulting matrix is also positive definite. And as per Part 1 of this exercise we know that the resulting matrix is positive if and only if $a > 0$ and $b - \frac{c^2}{a} > 0$ which is equivalent to $ab - c^2 > 0$.

1.3 Part 3

We can use what we just learned to show that the function $f(x, y) = 2x^2 + 3y^2 + 4xy$ will be positive or equal to 0 for all x and y . $\forall x, y \in \mathbb{R} : f(x, y) \geq 0$ is equivalent with saying that A is positive semi definite for some matrix A where $f_A = f$.

We can use the formula in section 3.7 to create the matrix for such function f_A :

$$A = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$$

Since A is symmetrical we can use the result of the previous part to show that A is positive definite. The conditions which have to be met is that $2 > 0$ and that $2 * 3 - 2^2 > 0$ both of which are true so A is positive definite and therefore positive semi definite.